

Onat Dogan Akca

+31 6 31 16 04 66 | onatakca@gmail.com | linkedin.com/in/onat-akca | onatakca.com

EDUCATION

University of Twente

Master of Science in Computer Science

- **Focus of Study:** Data Science and Machine Learning

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Science

Enschede, Netherlands

Aug. 2024 – Present

Champaign, IL

Aug. 2019 – May 2023

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, C, Bash

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, Hugging Face

MLOps & Cloud: Docker, Git, Google Cloud Platform (GCP), Weights & Biases, Linux

Concepts: Computer Vision, Deep Learning, Image Segmentation, Big Data Analytics

EXPERIENCE

Master's Research Intern – Machine Learning

Vlisco

Feb. 2026 – Present

Helmond, Netherlands

- Built an end-to-end motif segmentation system for textile design automation — **SAM3** exemplar-based detection with a custom topology-aware refinement model (**PyTorch**) — served by a dockerized async **FastAPI** + Redis GPU-worker service nearing production launch.
- Created the ground-truth benchmark solo: built a browser-based annotation tool and labelled ~380 designs into 261 motif units and ~18,400 instances.
- Trained topology-aware boundary refinement models (cIDice loss, automatically derived gap supervision), cutting boundary error (HD95) by **35%** and raising pixel-exact fill rate from 1.2% to **16.3%** on held-out designs.
- Benchmarked 12 methods on a custom evaluation harness (**PQ 0.512** vs. 0.452 raw SAM3) and built a **Photoshop UXP plugin** (React) replacing a ~40-minute-per-design manual workflow.

Graduate Research Assistant (Computer Vision)

Middle East Technical University (METU)

Oct 2023 – Sept. 2025

Ankara, Turkey

- Optimized **nnUNetv2** and **DeepLabv3** architectures for advanced eyelid gland segmentation, fine-tuning hyperparameters to maximize image analysis efficiency.
- Achieved a **DICE score of 0.84** for gland segmentation and **0.92** for eyelid segmentation, significantly surpassing existing benchmarks.

Software Engineer Intern

OpenProcessing.org

June 2022 – Aug. 2022

New York, NY (Remote)

- Engineered a collaborative filtering recommendation system using **Python** (Pandas, NumPy) to personalize content for a user base of **100,000+ monthly users**.
- Authored technical documentation and interactive **p5.js** tutorials for multiplayer game development.

Research Assistant (Deep Learning)

University of Illinois Chicago

May 2022 – Aug. 2022

Chicago, IL

- Designed deep neural networks in **TensorFlow** and **Keras**, with automated Python preprocessing pipelines, to extract survival patterns from crash-survivor sensor datasets.

PROJECTS

Süper Lig Football Analytics RAG | *Python, Transformers, Claude API, Streamlit*

Jan 2026 – Present

- Developed a bilingual RAG system combining semantic embeddings with BM25 keyword search using Reciprocal Rank Fusion for Turkish football Q&A.
- Implemented match-aware document chunking with automated entity extraction, deployed via Streamlit with Claude API integration for grounded, citation-backed answers.

EA75 Max Control | *Python, hidapi, radare2, pywebview, PyInstaller*

Jul 2026

- Reverse-engineered a keyboard's undocumented USB-HID protocol by disassembling its Windows-only vendor driver with radare2 and verifying every command on real hardware.
- Shipped a self-contained native macOS app enabling per-key RGB, TFT screen image/GIF uploads with per-page ACK flow control, and wireless power settings.

View full project portfolio at onatakca.com